



Using transfer learning model for brain tumor detection

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RESUMEN

Este trabajo investiga el uso de redes neuronales convolucionales y técnicas de aprendizaje por transferencia para clasificar imágenes de rayos X del cerebro, con especial atención a la detección de tumores. La base de datos utilizada contiene 253 imágenes de resonancia magnética, de las cuales 155 muestran tumores. La metodología incluyó el preprocesamiento de imágenes, como el recorte de bordes y la ecualización de histogramas, seguido del entrenamiento de un modelo VGG-16 previamente entrenado. Los resultados demostraron una exactitud del 93,35% en entrenamiento, lo que pone de relieve la eficacia del enfoque.

Palabras claves: CNN; Computación Visual; Transferencia de Aprendizaje; Tumor; VGG-16.

ABSTRACT

This work investigates using Convolutional Neural Networks and transfer learning techniques to classify brain X-ray images, focusing on tumor detection. The dataset contains 253 magnetic resonance images, of which 155 show tumors. The methodology included image preprocessing, such as edge cropping and histogram equalization, and training a pre-trained VGG-16 model. The results showed an accuracy of 93.35% in training, highlighting the approach's effectiveness.

Keywords: CNN; Computer Vision; Transfer Learning; Tumor; VGG-16.

1 Introduction

According to Pinho Costa et al. (2024), cancer is a term that encompasses more than 100 diseases, all characterized by the uncontrolled growth of cells that invade tissues and organs. The National Cancer Institute (INCA 2024a) states that these cells have an accelerated and usually aggressive division process, resulting in the formation of tumors. The

origin of the affected cells defines different types of cancer. For example, when cancer originates in tissues such as the skin or mucous membranes, it is classified as carcinoma. In contrast, it is called sarcoma when it arises in bones, muscles, or cartilage (INCA 2024b).

In 2022, approximately 18 million new cancer cases were diagnosed worldwide, of which around 322,000 were brain cancer, resulting in approxi-